

Integrity Algorithm of a Tightly-coupled KF based GNSS/IMU Navigation System with Redundant IMUs

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Recently, autonomous vehicles, including autonomous drones or self-driving cars, are getting a great deal of attention with the expectation that they can partially replace or augment human performance while improving operational efficiency and increase the range of applications. Since those vehicles conduct control and path-following operations autonomously by trusting the given navigation information, the safety of autonomous vehicles predicated on a sufficient level of navigation integrity.

Recent studies suggest that the required level of navigation integrity of autonomous vehicles should be set comparable to the navigation integrity mandated in civil aviation, which ranges from 10^{-6} to 10^{-9} depending on applications [1]. In response, researchers leverage the concept of navigation integrity, carried out in civil aviation when developing integrity algorithms of autonomous vehicle navigation systems.

One most key research transition when developing the integrity algorithm of autonomous vehicle navigation compared to those of civil aviation is to consider various multiple navigation sensors and different kinds of filtering algorithms. Currently, multiple sensors, including Global Navigation Satellite System (GNSS), Inertial Measurement Unit (IMU), vision sensor, barometer, magnetometer, and LIDAR, are integrated to provide the navigation solution of autonomous vehicles. As the many inputs from multiple sensors are integrated, navigation solutions become more vulnerable to potential sensor faults. Potential integrity risks from all the sensors used for the system should be considered along with its integration algorithm to assure integrity of a multi-sensor navigation system.

While there could be various combinations of sensor integration scenarios, a tightly-coupled Kalman filter (KF)-based GNSS/IMU navigation system is generally utilized as a basis for typical multi-sensor navigation systems. Since the KF is a recursive estimator, a recursive sensor fault impact on KF state estimates should be considered when quantifying integrity risks. The KF based integrity algorithms that quantify the recursive fault impact on state estimates are actively being discussed in recent publications [2]-[4]. These previous studies dealt with sensor faults that occur in the GNSS measurement sequence, which is applied for the KF measurement update, assuming IMU is a fault-free sensor.

However, studies reported that IMU is also vulnerable to potential faults, and the reliability of an inertial sensor is critically vital for operations [5]. When GNSS and IMU are integrated with a KF, IMU sensor biases are generally augmented in a KF state vector and are continuously estimated from the filter based on an error dynamics model. In a closed-loop KF algorithm, the estimated IMU biases for each axis are fed back and subtracted from an input IMU measurements at the next epoch. However, this bias estimation process is not enough for capturing IMU faults because 1) IMU faults may not follow the pre-defined error dynamics model, 2) even if we assume that IMU faults follow the pre-defined error model, there is a time delay to perfectly capture the faults.

Thus, the integrity algorithm for the IMU faults occurring in the state prediction step is also required in multi-sensor navigation systems to fully assure the system safety, based on the fact that there are two main filtering steps in KF, which are a state prediction and measurement update.

In our previous work [6], we illustrated how an unexpected step type IMU faults (potentially due to fracture, stiction, or delamination) affect the KF estimates when it is used for the state prediction in a GNSS/IMU integrated navigation system. It was also shown that the fault impact on KF state estimates due to the IMU sensor faults could be expressed using the KF innovation vector in real-time, whereas it is not applicable in case of GNSS sensor faults.

In our following work [7], we proposed two protection levels (PLs) determined for the IMU sensor fault hypothesis using the KF innovation vector and additional noise bounding terms. One is “a real-time $PL_{KF,IMU}$ ” which supports position error bounding in real-time by accounting for actual fault and noise impact on user position, and the other is “a predictive $PL_{KF,IMU}$ ” which can be used for availability evaluation before initiating the operation based on the determined worst-case position error after fault monitoring. Based on the two PL concepts as well as PLs for nominal condition and GNSS fault hypothesis, operation concept was suggested both prior to operation and in a real-time operation against the IMU sensor fault hypothesis.

Through the studies in [6] and [7], we established an integrity architecture of tightly-coupled GNSS/IMU navigation systems against all sensor faults, including the IMU sensor fault hypothesis. However, the integrity architecture was preferentially designed by assuming that only a single IMU is utilized for the system without redundancy. Thus, a single IMU sensor failure was assumed to guarantee the integrity risk probability assigned to the IMU failure hypothesis when designing an integrity fault tree, a fault monitor, and a PL equation. The critical limitation of not having redundant IMU sensors is that sensor exclusion is meaningless in the case of IMU sensor fault because no further sensor utilization is possible, and navigation integrity of the integrated navigation system is not guaranteed after IMU exclusion.

Currently, most autonomous navigation systems utilize multiple redundant IMU sensors to improve robustness. There have been many studies on fault detection and isolation algorithms in the presence of redundant IMU sensors. However, algorithms that quantify the integrity risk considering the recursive IMU fault impact on user position within the KF in the presence of redundant IMUs have not been investigated. Thus, it is required to employ the developed integrity algorithm to realistic operational conditions where there are available redundant IMU sensors integrated with GNSS.

This paper extends the concept of integrity algorithm for a tightly-coupled KF-based GNSS/IMU integrated navigation system by accounting for multiple redundant IMU sensors. The integrity architecture is capable of IMU sensor fault detection and exclusion while assuring total system integrity. Three filter design scenarios in the presence of multiple redundant IMU sensors [8] are considered which are

- (a) an independent filter that utilizes a state estimates from a single set of a filter at a time, and uses other filters as backup solutions,
- (b) a centralized filter which utilizes all the IMU sensor measurements by stacking in a single matrix to compute a state estimates, and
- (c) a federated filter (or known as a decentralized filter) where all the parallel local filters process each IMU sensor measurements and local estimates are subsequently fused in a master

filter to achieve the global estimations.

For each filtering scenarios, we modified the integrity architecture with the consideration of following three key design processes which are,

- (1) design of integrity architecture including integrity risk allocations for each fault hypothesis, each sub-filters, and corresponding fault monitoring algorithms,
- (2) development of fault detection and exclusion logic and algorithm
- (3) computation of PLs considering filtering process against all sensor fault hypothesis which are a nominal condition, a GNSS fault condition, and an IMU fault condition

An integrity algorithm for the independent filter (a) is a simple extension of the previously developed algorithm with additional redundant parallel filters so that modification on the risk allocations, fault detection, and PL computation algorithms are not necessary. However, we can take advantage of parallel filters to ensure integrity in the event of an IMU sensor failure without availability interruption.

In case of both the centralized filter (b) and federated filter (c), modifications are required with careful consideration due to reasons including 1) total integrity allocated to an IMU fault hypothesis should be divided and be taken from multiple redundant IMU sensors, 2) the form of fault detection algorithm should be modified due to changed filtering architecture, and logic for sensor exclusion after fault detection should be established, 3) PL computation algorithm should be modified in line with the filtering architecture with redundant IMU sensors, 4) lastly, correlation between sub-filters should be considered in case of the federated filter to assure pre-defined total integrity risk.

Based on the extended integrity algorithm for the KF-based GNSS/multiple redundant IMUs navigation systems, we suggested an operational concept for each filtering scenario utilizing the PLs which are outputs of the algorithm. In addition, we compared the PLs for each filtering scenarios. The proposed integrity algorithm can be widely used to fully ensure the integrity as well as robustness of autonomous vehicles.

What is new and innovative?

- Proposed an integrity architecture of a tightly-coupled KF based GNSS/IMU navigation system considering redundant IMU sensors
- Developed integrity algorithms that consist of risk allocation, fault monitor and exclusion, and protection level computation for three filtering scenarios in the presence of redundant IMU sensors, which are (1) an independent filter, (2) centralized filter, and (3) federated filter
- Verified and showed the applicability of the developed integrity algorithm, and compared protection levels between filtering scenarios

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